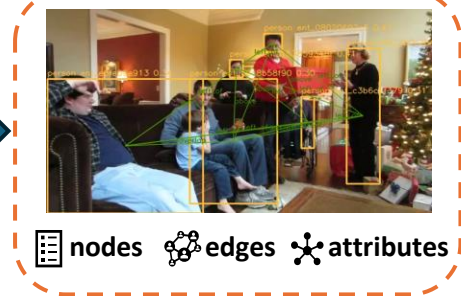


Video (Keyframe Extraction)



Initial Scene Graph



Human Feedback

✗ ✓ - approve / reject
🔍 ↺ - provide correction

Final Scene Graph



Hierarchical CYCLE Verification

Claims Probe Generator

object: person_1
label: person
relation: person_1 left_of person_2
attribute: person_1 is wearing red

Level 1

Local:
object / attribute / spatial relation

Claim: person_1 is wearing red
Reasoning: detected red region on upper body with high confidence
Vote: ✓
Score: 0.92
.....

Level 2

Temporal:
consistency across frames

Claim: person_1 remains left of sitting across frames
Reasoning: - Spatial relation (Q1, Q2) - Posture is inconsistent time (Q3)
Vote: ✓
Score: 0.85
.....

Level 3

Global:
overall scene coherence

Claim: The scene graph is globally consistent with the caption
Reasoning: -Conflict detected: person_1 is labeled as wearing red, but visual evidence is weak
Vote: ✗
Score: 0.40

Graph Update

- update labels
- update relations
- update bounding boxes

Aggregation

Agent	Score
Local	+ 0.92
Temporal	+ 0.85
Global	- 0.40
Final Score: 0.50 →	